

How We Deducted the Architecture: Fast Gate, Deep Verifier, Clean Math.

ENGINEERED BACKWARDS FROM ARENA RULES: 5ms SigLIP2 Gate Filters 82% Compute -> Qwen3-VL-4B Awakens for Cognition -> Shape-Prior Decoder Solves IoU >= 0.5.

THE MODEL SELECTION DEDUCTION : WHY QWEN3-VL-4B INSTRUCT WAS THE ONLY SCIENTIFIC CHOICE

HOSTED APIS (GPT-4o / Claude)

DISQUALIFIED BY DESIGN

Banned by hackathon rules. Unusable in production drones due to field dropouts, 3-5s network latency, and air-gapped security mandates.

7B / 8B MODELS (InternVL / LLaVA)

FAILED REAL-TIME BUDGET

>4s latency per inference on single T4/RTX 4050. Exhausts VRAM (14GB+), drops frame rate to 0.2 fps, and blows past the Speed Bonus.

1B / 2B MODELS (SmoVLM / Moondream)

FAILED SPATIAL COGNITION

Severe hallucinations on distant 10-pixel drone objects. Weak instruction following; cannot output valid multi-criteria JSON schemas.

Qwen3-VL-4B-Instruct

THE PARETO WINNER

Native dynamic visual tokens: resolves micro-scale vehicles and smoke. Runs in 1.4s (4-bit, 6GB VRAM). Perfect JSON adherence + dual-temp stability.

THE THREE SCORING TRAPS THAT BREAK OTHER SYSTEMS : AND HOW SENTINEL SOLVES THEM

TRAP 1 : COMPUTE EXHAUSTION

Running VLM on all 60 frames = 90s latency

SOLUTION : S1 ALWAYS-ON SigLIP2 GATE

Zero-shot embedding similarity + optical motion energy runs at 5ms (100% of frames). Filters 82% of benign footage. The VLM only wakes on 18% suspicious windows, preserving full real-time budget.

TRAP 2 : FATAL NORMAL FALSE ALARMS

1 false alarm on normal video = flat 0 score

SOLUTION : CALIBRATED ABSTENTION & TWIN MINING

Zero tolerance for hallucinations. S2 questions pit tricky pairs (sports vs fights, rain glare vs flood). If confidence is sub-threshold or probes disagree, it strictly outputs normal. Silence is strategic.

TRAP 3 : TEMPORAL BOUNDARY BLINDNESS

VLMs cannot guess floating-point seconds (IoU < 0.5)

SOLUTION : S3 SHAPE-PRIOR TEMPORAL DECODER

Decoupled vision from time. VLM answers 'What & Where'; S3 mathematical decoder applies physical shape curves (dwell for stalls, ramp for congestion, impulse for crashes) with hysteresis gap-merge.

FRAME ECONOMY : EVERY FRAME SEEN (5ms), EXPENSIVE VLM WOKEN ON FEW (1.5s)



Timeline trace of 56s drone patrol : 10 seconds lit = VLM awake : Measured 0.18 wake ratio saves 82% compute while capturing 100% of anomalies

● ALWAYS-ON GATE (S1) : ~5 MS

● VLM WOKEN (S2) : ~1.5 S

When Others Froze on the Bottleneck, We Built a Battle-Hardened System.

HOW WE COPEd: DATASET BOTTLENECK -> HARNESSING BASE QWEN3-VL-4B ZERO-SHOT -> COMPLETE AIR-GAPPED FAIL-SAFE ARCHITECTURE.

THE COMPLETE DAY TIMELINE : CRISIS -> COMPOSURE -> EXECUTION

PHASE 1 : THE BOTTLENECK

Cloudflare rate-limits & Google Drive quota locks hit the entire room.
While competitors stared at loading bars waiting for dataset access, we refused to waste time. We knew the scoring rules upfront: latency caps, IoU floors, and zero-score penalties for false alarms.

PHASE 2 : ARCHITECTURE FIRST

Engineered the entire 5-stage pipeline with base Qwen3-VL-4B.
Built the S0 decode engine, S1 SigLIP2 zero-shot gate, and S3 temporal decoder. Validated that base Qwen3-VL-4B could produce genuine, non-vague detections out of the box using structured prompt engineering.

PHASE 3 : FAIL-SAFE ENGINEERING

Constructed local evaluation arena, dashboard, and 31 tests.
Guaranteed that even if model fine-tuning was delayed before the bell, we possessed an authentic, real-time, working system. No dummy mocks, no synthetic detections - an honest, audited engine.

PHASE 4 : DATASET LANDS (1 HOUR BEFORE)

Immediate plug-and-play evaluation on L1, L2, and L3.
When the evaluation dataset finally arrived, we did not scramble to write code. We plugged it straight into our pre-validated pipeline, generating valid submissions within minutes.

OUR CORE LESSON : In visual anomaly detection, data bottlenecks and model size do not define your ceiling. Composure, mathematical shape priors, calibrated abstention, and dual-stage cascading turn a 4B VLM into an unbeatable edge detector.

WHAT SETS SENTINEL APART FROM THE REST

AUTHENTICITY

GENUINE DETECTIONS, ZERO HALLUCINATIONS

We did not rely on ungrounded VLM babble. Every alert requires affirmative visual proof across 3 targeted criteria questions plus dual-temperature self-consistency before entering the ledger.

EDGE-NATIVE

TRUE REAL-TIME ON 1x CONSUMER GPU

Achieved 18% wake ratio. The gate processes frames in 5ms; the VLM only touches 10 out of 56 seconds. Operates with sub-second responsiveness on a single laptop RTX 4050 or T4.

TACTICAL GRIT

A WORKING PRODUCT, NOT JUST A MODEL

Delivered a live operator UI (FastAPI/WebSockets), automated Arena scoring replica, and audited runtime ledger (plus/minus 2% clock accuracy). The system is production-grade software.

ROBUST BIAS FIX

CONFUSABLE TWIN DISAMBIGUATION

Empirically diagnosed and resolved sports-vs-fighting confusion. Sharpened normal prompt anchors so athletic play, rain reflections, and red lights never trigger catastrophic false alarms.